

ONLINE LAW FIRM MANAGEMENT SYSTEM

Software Requirements Specification, System Design, and Database Documentation

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1. Introduction

The Online Law Firm Management System (OLFMS) is a secure web-based application for managing client legal-service requests, cases, appointments, lawyer assignments, and case files. It supports a workflow in which a client registers and submits a case request, a company manager reviews the request and appointment, and an appropriate lawyer is assigned to handle the case.

1.1 Purpose

The system centralizes routine law-firm activities, reduces reliance on informal communication and paper records, and gives clients controlled access to their own case information and service guidance.

1.2 Scope

The first release covers authentication, password recovery, document upload/download, case review, appointment booking, case assignment, role-based access, and a client user guide. It does not include online payments, court filing integration, billing, or video conferencing unless added in a later release.

1.3 Stakeholders

Stakeholder	Interest / responsibility
System administrator	Administers users, roles, security settings, and system operation.
Company manager	Reviews client case orders and appointments; assigns lawyers to cases.
Lawyer	Reviews assigned cases, client details, appointments, and relevant files.
Client	Requests legal assistance, books appointments, uploads/downloads files, and views own case details.

2. Requirements Specification

2.1 Functional Requirements

- The system shall allow new client to register an account by using (Full name, password, Email, Location, Contacts, Gender)
- The system shall allow users to login by using (username(Email) and password)
- The system shall allow manager to register new lawyers by using (Full name, Email, Education level, Phone number and location)
- The system shall provide “forgot password” option
- The system shall allow Lawyer to upload law and constitutional document files (.pdf, .jpg)
- The system shall allow client to upload case related document files (.pdf, .jpg)
- The system shall allow users to download files (.pdf, .jpg)
- The system shall allow client to book appointment with lawyer
- The system shall allow user to review case details
- The system shall enforce law based access control
- The system shall allow client to view system user guide
- The system shall allow user to logout
- The system shall record number of login attempts

2.2 Non-Functional Requirements

- The system shall enforce password rule of minimum of 8 characters including number, special characters and at least upper case letter and lower-case letter
- The system shall encrypt password by using hash function
- The system shall lock out after 3 failed attempts
- The system shall return results within 2 seconds
- The system shall be available to all system users whenever needed
- The system shall allow multiple users at the same time
- The system shall provide a shall provide a simple, clear and responsive interface when viewed in Personal computer, mobile phone, Tablets and desktop
- The system shall limit maximum of 10mb per uploaded file
- The system shall validate user inputs, enforce database constraints, prevent invalid and avoid duplicates
- The system shall support modern browsers such as google chrome , Firefox, safaris.
- The system shall store uploaded files in secure file directory
- The system shall send a password reset link valid for 30 seconds

3. Roles and Permissions

Access control must use role-based access control (RBAC) and must enforce ownership: a client may access only their own cases and files, while a lawyer may access cases assigned to them.

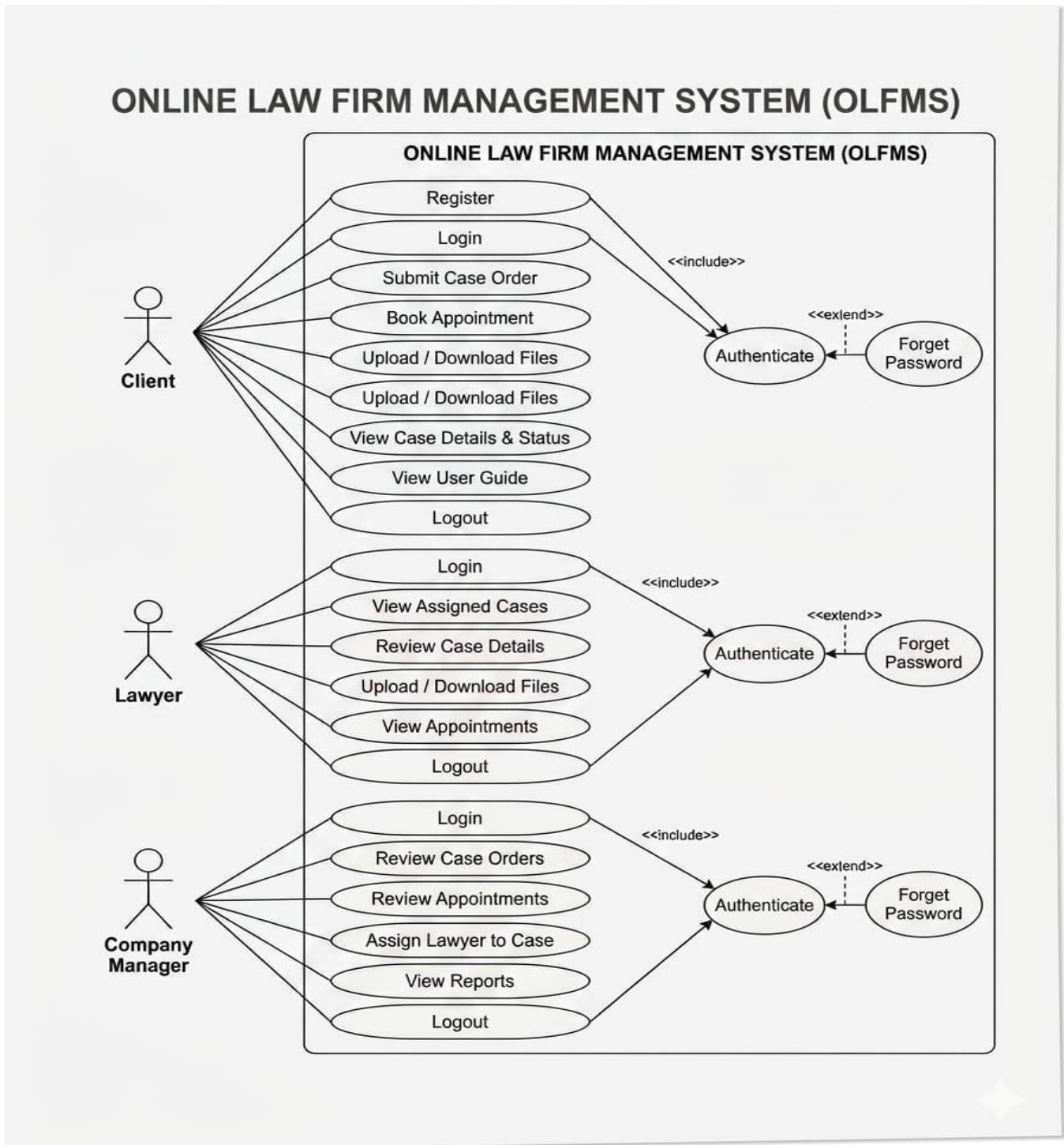
Function	Administrator	Manager	Lawyer	Client
Register / login / logout	Yes	Yes	Yes	Yes
Manage users / roles	Yes	No	No	No
Submit case order	No	No	No	Yes
Review case details	All (admin purpose)	All managed cases	Assigned cases	Own cases
Assign lawyer	Optional oversight	Yes	No	No
Book appointment	No	Review	View assigned	Yes
Upload / download files	Authorized support	Authorized cases	Assigned cases	Own cases
View user guide	Yes	Yes	Yes	Yes
Manage profile	Yes	Only password	Only password	Only password

4. Process Overview

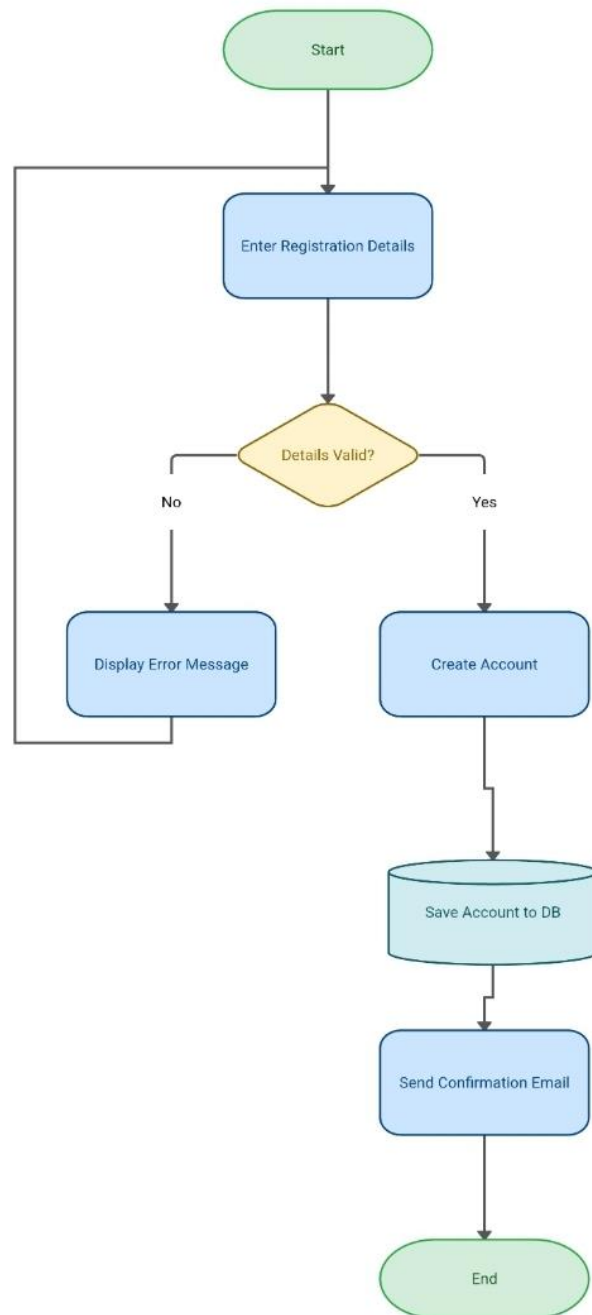
1. A client registers, logs in, and creates a case order containing a title, description, category, and optional evidence files.
2. The company manager reviews the submitted case and requested appointment.
3. The manager assigns an available lawyer.

4. The lawyer reviews the assigned case and client details.
5. The client follows case status, accesses permitted files, and uses the user guide as needed.

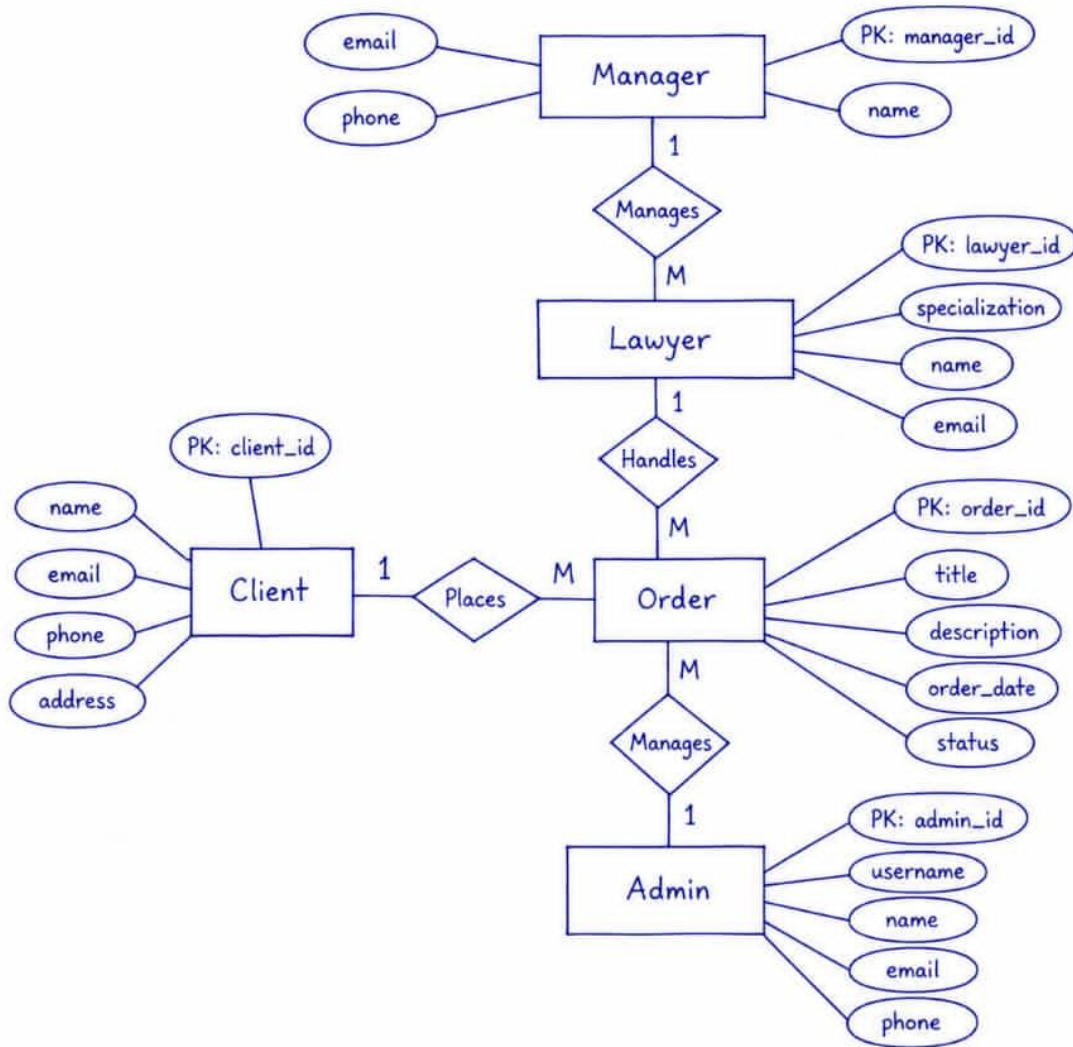
5. Use Case Diagram



6. Activity Diagram

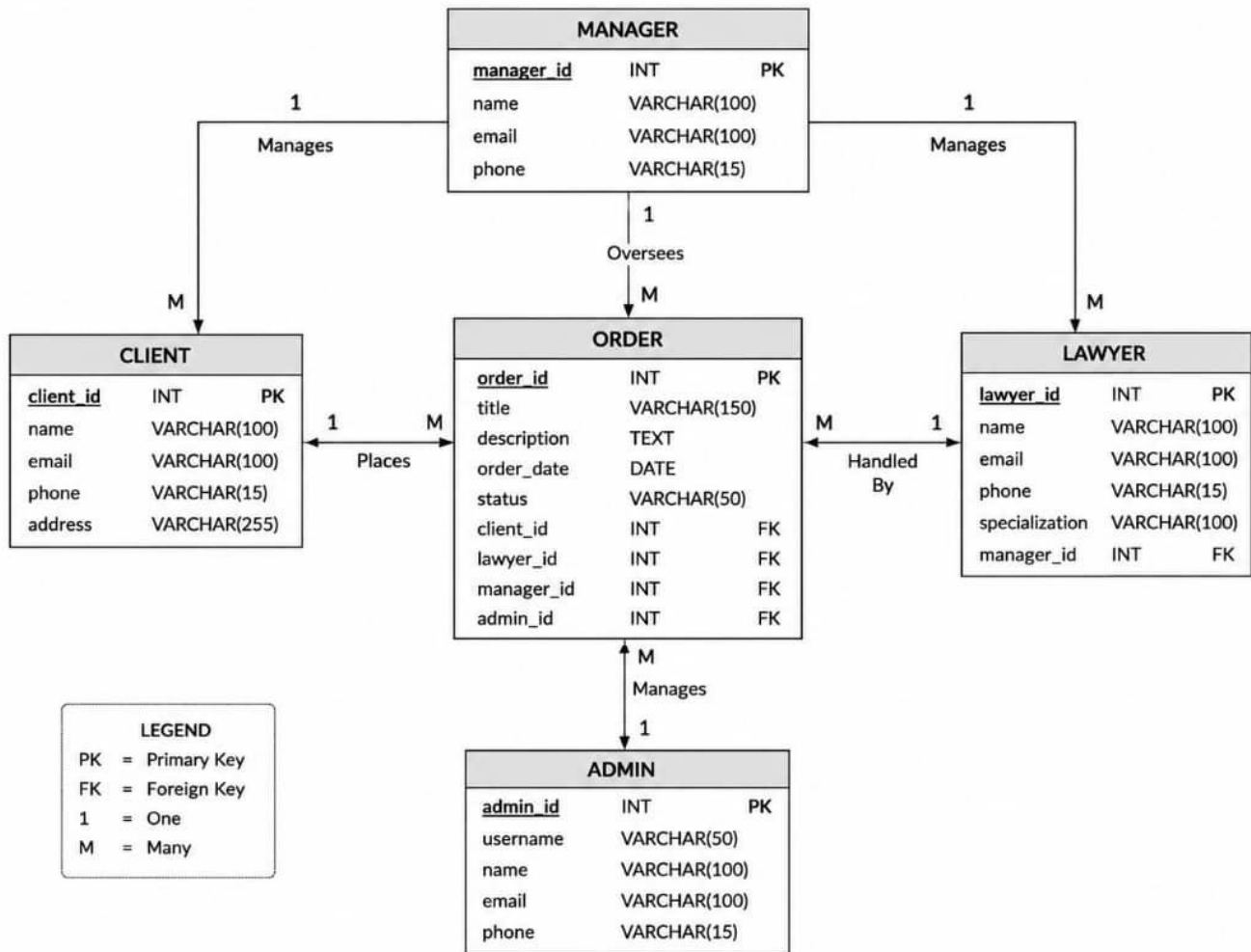


7. Entity-Relationship Diagram



8. Logical Database Schema

LOGICAL SCHEMA – ONLINE LAW FIRM



9. Security and Validation Design

Authentication: accept a username and password only over HTTPS, compare the submitted password with the stored adaptive hash, create a secure server-side session or signed token, and invalidate the session on logout. Increment the failed-attempt counter on every invalid login; lock the account after the third consecutive failed attempt.

Password reset: generate a cryptographically secure random token, store only its hash, mark it single-use, and reject it after 30 seconds. Do not disclose whether a supplied email or username exists; display a generic confirmation message.

File handling: validate extension and actual content type; permit only PDF and PNG; reject files larger than 10 MB; generate a server-side storage key rather than using the supplied filename; scan files where infrastructure permits; and authorize every download using the requesting user, role, ownership, and case assignment.

Privacy: case data and uploaded material are sensitive. Apply least-privilege access, audit access to files and cases, encrypt data in transit, maintain backups securely, and establish retention/deletion rules appropriate to the firm's legal and regulatory obligations.

10. Test Cases

Test ID	Scenario	Expected result
TC-01	Register with valid client information and a password meeting policy.	Account and client profile are created; client can log in.
TC-02	Register using duplicate username.	Registration is rejected with a uniqueness validation message.
TC-03	Attempt login with wrong password three times.	Account is locked after the third failed attempt; access is denied.
TC-04	Request and use reset link within 30 seconds.	Password is changed after valid token verification; token becomes unusable.
TC-05	Use reset link after 30 seconds.	System rejects token as expired.
TC-06	Upload a 9 MB PDF to own case.	File is stored securely and is visible to authorized users.
TC-07	Upload an 11 MB PNG or an unsupported file type.	Upload is rejected without storing the file.
TC-08	Client tries to open another client's case/file.	System denies access and records the attempt if auditing policy requires.
TC-09	Manager assigns a lawyer to a submitted case.	Assignment record is created, case status changes, and lawyer can view the case.
TC-10	Measure common dashboard/case-view request under expected load.	Result is returned within 2 seconds.

11. Deployment and Acceptance

Recommended architecture: a browser-based client application, an application/API service containing business logic and RBAC checks, a relational database, a secure private file store, and an email service for reset links and notifications. Separate development, testing, and production environments; do not use production case records for testing.

The project is accepted when all functional requirements are demonstrable, all listed security controls pass their test cases, role and ownership checks prevent unauthorized access, uploads meet file restrictions, reset links expire after 30 seconds, the three-attempt lockout works, and agreed performance testing shows normal results within 2 seconds.